Rethinking case attraction on Ancient Greek infinitive clauses

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Example

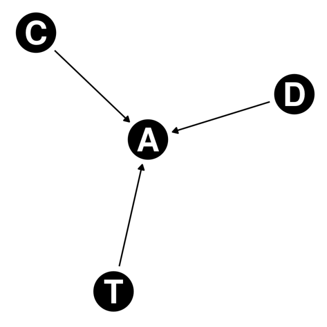
(1) a. συμβουλεύει **τῷ Ξενοφῶντι ἐλθόντα** είς Δελφοὺς ἀνακοινῶσαι τῷ θεῷ περὶ τῆς πορείας.
He advices Xenophon to go to Delphi and ask the god about the travel. (Xen. *Anab.* 3.1.5)

b. ἀφῆκε **μοι ἐλθόντι** πρὸς ὑμᾶς λέγειν τἀληθῆ.
He allowed me to go and speak the truth in front of you. (Xen. *Hell.* 6.1.13)

Direct Acyclic Graph: Agreement

Framework: Corbett (2006)

Features of the **C**ontroller, **T**arget and **D**omain “cause” non-canonical **A**greement



Domain of attraction: Part of speech and copular infinitive

(2) Agreement and Predicate Hierarchies (Corbett 2006, p. 233):

Noun

↗

Adjective

↗

Participle

↗

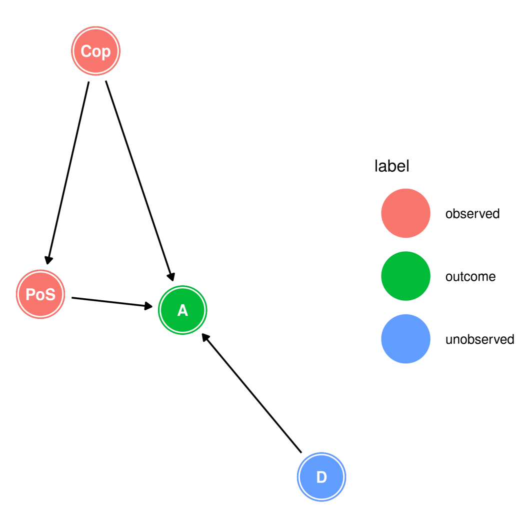
Attributive → Predicative → Relative Pronoun → Personal Pronoun

Verb

- ⟵⟶ +

Likelihood of agreement with semantic justification

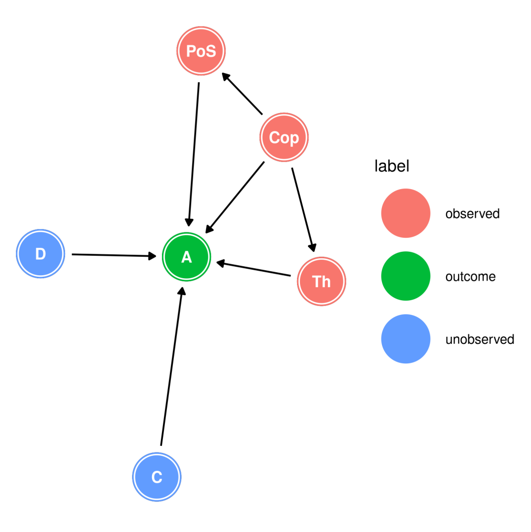
The domain of attraction: DAG



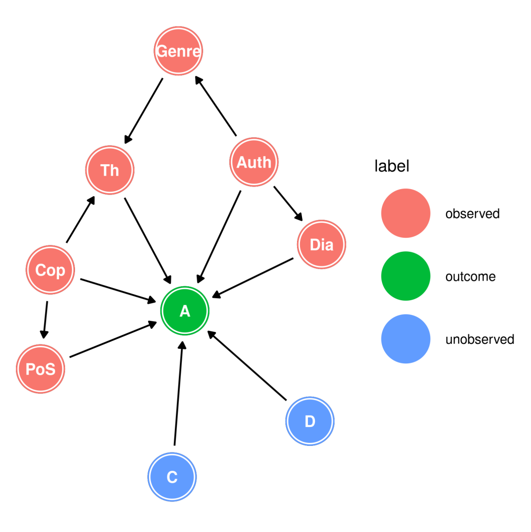
The controller of attraction: Thematic role

(3) ἀλλ᾽ ἐξαρκῇ **σοι** τὰ ἔργα αὐτῶν **ὁρῶντι** σέβεσθαι καὶ τιμᾶν τοὺς θεούς.
But it is enough for you to look at these deeds and respect the gods. (Xen. *Mem.* 4.3.13)

The controller of attraction: DAG



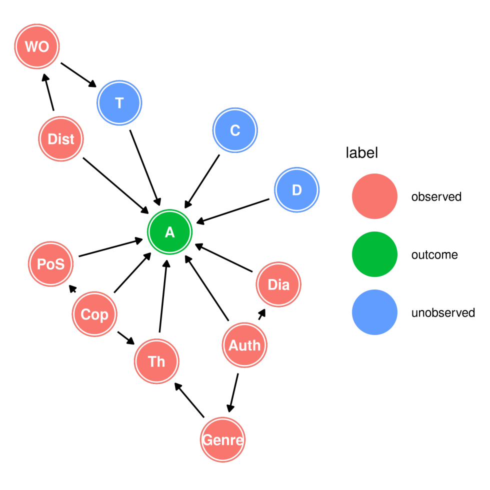
Issues of dialect and authorship: DAG



The target of attraction: distance and precedence

(4) ὡς **ἀγαθοῖς** τε **ὑμῖν** προσήκει εἶναι.
That you may be good people. (Xen. *Anab.* 3.2.11)

The target of attraction: final DAG



Tooling: R (R Core Team 2013), the packages `tidyverse` (Wickham et al. 2019), `dagitty` (Textor et al. 2016), `ggdag` (Barrett 2025), and `rethinking` (McElreath 2020), as well as the software Stan (Stan Development Team 2025a,b).

Generalized linear model

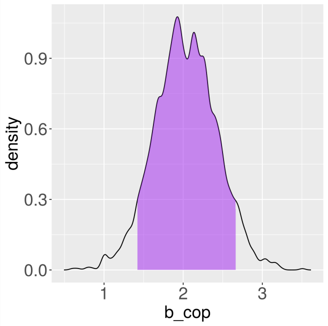
$$
\begin{aligned}
\mathsf{A} &\sim \mathsf{Bernoulli}(p_i) \\
\mathsf{logit}(p_i) &= \alpha + \beta_{\mathsf{Cop}}\mathsf{Cop}_i + \beta_{\mathsf{Dist}}\mathsf{Dist}_i + \beta_{\mathsf{WO}}\mathsf{Prec}_i \\
&+ \beta_{\mathsf{PoS}} \sum_{j=0}^{PoS_i - 1} \delta_{j_{\mathsf{PoS}}} + \beta_{\mathsf{Th}} \sum_{j=0}^{Th_i - 1} \delta_{j_{\mathsf{Th}}} \\
&+ \beta_{\mathsf{Auth}_i} + \beta_{\mathsf{Dia}_i} \\
\alpha &\sim \mathsf{Normal}(0, 1) \\
\beta_{\mathsf{Cop}}, \beta_{\mathsf{Dist}}, \beta_{\mathsf{Prec}} &\sim \mathsf{Normal}(0, 1) \\
\beta_{\mathsf{PoS}}, \beta_{\mathsf{Th}} &\sim \mathsf{LogNormal}(0, 1) \\
\delta_{\mathsf{PoS}}, \delta_{\mathsf{Th}} &\sim \mathsf{Dirichlet}(2) \\
\beta_{\mathsf{Auth}} &= z_{\mathsf{Auth}} * \sigma_{\mathsf{Auth}} \\
\beta_{\mathsf{Dia}} &= \overline{\beta}_{\mathsf{Dia}} + z_{\mathsf{Dia}} * \sigma_{\mathsf{Dia}} \\
\overline{\beta}_{\mathsf{Dia}}, z_{\mathsf{Auth}}, z_{\mathsf{Dia}} &\sim \mathsf{Normal}(0, 1) \\
\sigma_{\mathsf{Auth}}, \sigma_{\mathsf{Dia}} &\sim \mathsf{Exponential}(1)
\end{aligned}
$$

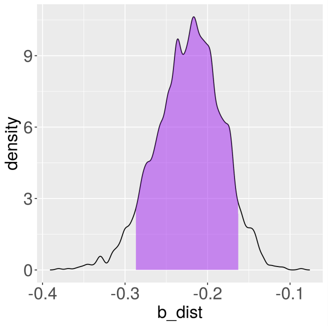
A word on Ordered Categorical Predictors

PoS	Cummulative effect
Participle	$\beta_{\text{PoS}} * 0$
Adjective	$\beta_{\text{PoS}} * (\delta_{\text{Adj}})$
Noun	$\beta_{\text{PoS}} * (\delta_{\text{Adj}} + \delta_{\text{Noun}})$

Copula and Distance

Effect	mean	sd	5.5%	94.5%	rhat
β_{Copula}	2.03	0.39	1.42	2.66	1
β_{Dist}	-0.22	0.04	-0.29	-0.16	1





Part of Speech and the Predicate Hierarchy

Effect	mean	sd	5.5%	94.5%	rhat
β_{PoS}	0.45	0.28	0.11	0.95	1
δ_{Adj}	0.53	0.18	0.23	0.82	1
δ_{Noun}	0.47	0.18	0.18	0.77	1

PoS	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Participle	$\beta_{\text{PoS}} * 0$	0.00	0.00	0.00	0.00	NA
Adjective	$\beta_{\text{PoS}} * \delta_{\text{Adj}}$	0.25	0.18	0.05	0.60	1
Noun	$\beta_{\text{PoS}} * (\delta_{\text{Adj}} + \delta_{\text{Noun}})$	0.45	0.28	0.11	0.95	1

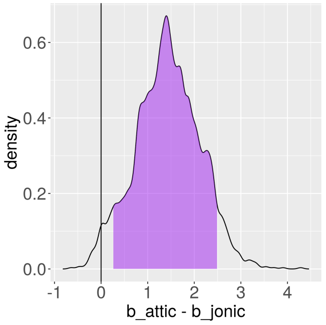
Thematic Role of the Controller

Effect	mean	sd	5.5%	94.5%	rhat
β_{Theta}	2.47	0.40	1.84	3.12	1
δ_{Exp}	0.34	0.11	0.17	0.51	1
δ_{Agent}	0.66	0.11	0.49	0.83	1

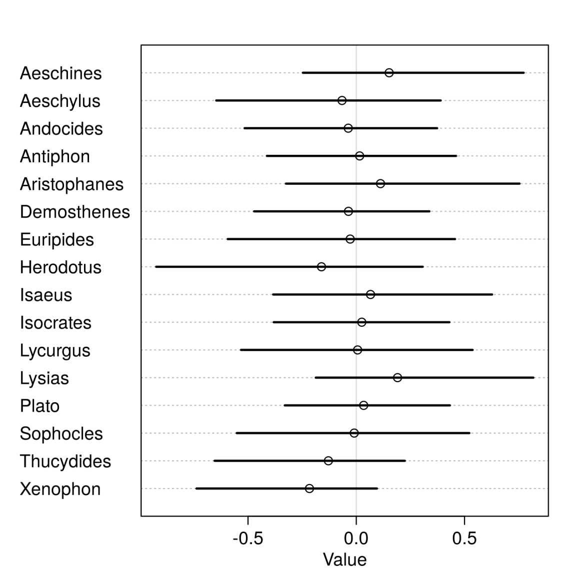
Theta	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Recipient	$\beta_{\text{Th}} * 0$	0.00	0.0	0.00	0.00	NA
Experiencer	$\beta_{\text{Th}} * \delta_{\text{Exp}}$	0.83	0.3	0.38	1.34	1
Agent	$\beta_{\text{Th}} * (\delta_{\text{Exp}} + \delta_{\text{Agent}})$	2.47	0.4	1.84	3.12	1

Dialect

Effect	mean	sd	5.5%	94.5%
$\overline{\beta}_{\text{Dia}}$	-0.49	0.84	-1.78	0.90
σ_{Dia}	1.20	0.78	0.30	2.67
β_{Attic}	-0.16	0.89	-1.55	1.28
β_{Jonic}	-1.59	1.01	-3.19	-0.02
$\beta_{\text{Attic}} - \beta_{\text{Jonic}}$	1.43	0.69	0.26	2.49



Authorship


Aeschines
Aeschylus
Andocides
Antiphon
Aristophanes
Demosthenes
Euripides
Herodotus
Isaeus
Isocrates
Lycurgus
Lysias
Plato
Sophocles
Thucydides
Xenophon
-0.5
0.0
0.5
Value

Thank you!

Contact:

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Data and code available at

`github.com/caiogeraldes/icagl2025`